TinyStar’Ex

The astronomical spectrograph you build yourself

buil.astrosurf.com/TSE

TinyStar’Ex is a high-performance astronomical spectrograph, with its own built-in pointing system, that you build yourself. The component kit is available from the shop; you print the mechanical parts and assemble them by following detailed instructions. It is simple, enjoyable, and economical. Compatible with the cameras commonly used for deep-sky imaging (not supplied), TinyStar’Ex suits the curious newcomer to spectrography and the experienced observer alike. You learn as you build it, and then as you use it.

TECHNICAL SPECIFICATIONS

Grating	300 lines/mm, interchangeable for future applications
Slits	4 selectable widths: 14, 20, 26, and 32 µm
Optics	apochromatic doublets in low-dispersion glass
Focal lengths	collimator 50 mm, camera 50 mm
Aperture	telescopes as fast as f/4
Spectral range	375 to 840 nm in a single exposure (IMX585 sensor), extendable to 1,050 nm with an order-sorting filter (not supplied)
Resolution at 656 nm	0.84 nm (R = 780) with the 14 µm slit; 1.2 nm (R = 550) with the 20 µm slit
Main camera	M42 × 0.75, back focus 17.5 mm
Pointing / guiding	31.75 mm (1.25-inch) barrel
Assembly	design that minimizes adjustments
Mass	700 g without camera





RECOMMENDED CAMERAS

Two cameras are needed to get the most out of TinyStar’Ex.

- **A main camera**, for recording the spectra. Any model with a back focal length of 17.5 mm and an M42 × 0.75 (T2) interface will do. A cooled camera with an IMX585 sensor is ideal, thanks to its small pixels and its format. If you own another model, do not worry: it will at least let you take your first steps in astronomical spectrography, and very probably much more. Even a colour camera can be used. The first step is what counts!
- **A pointing and guiding camera**, to place a star or any other object on the slit and keep it there during the exposure. Conventional ‘stick’ guide cameras or entry-level planetary cameras are suitable.

THE CREATORS

STAROS Projects

TinyStar’Ex is designed by the **STAROS Projects** team, already behind the SUNSCAN spectroheliograph. It is a direct descendant of Star’Ex, conceived by Christian Buil.

RECOMMENDED TELESCOPES

- TinyStar’Ex performs at its best on telescopes of modest size, from 80 to 250 mm in diameter.
- It can be used on larger apertures, at the cost of a noticeable loss of spectral resolution, since the 26 or 32 µm slits must then be used. In return, the instrument becomes remarkably effective for applications where the signal-to-noise ratio matters more than resolution, such as measuring the spectral energy distribution (SED) or identifying very faint sources.
- In spectrography, reflecting telescopes are preferable to refractors. A mirror brings all wavelengths to the same focus, whereas a lens suffers from chromatic aberration, particularly in the blue. This is also true of so-called apochromatic refractors as soon as you move towards the ultraviolet. A refractor remains usable, at least for getting started, but TinyStar’Ex reveals its full power behind a reflector.
- A major strength of TinyStar’Ex is its ability to work with very fast telescopes, down to f/4, with virtually no loss of signal. Its preferred range is f/4 to f/8, but it remains versatile: with a Cassegrain or a Maksutov, for example, you will also obtain good spectra.